



#EI

#Discovery:

Something that exists in nature, but has been found by someone for the first time.

Eg: Columbus → America, Newton → gravity

#Invention: creation of a new entity with the help of principles and theories already present

Eg: Washing machine, fridge, etc

#Innovation: Repurposing / Modifying an invention

Eg: Frothing of milk by steam for coffee ☞

#Jugad (Frugal invention):

- Indian's have always been inventors, innovators, researchers, etc
- There was a time when India was looted for its resources.
- But no matter how much resource crunch one may have there is a way to survive and live a better life,
- This can be done by repurposing things, modifying stuff to use them for purpose maybe no one would have thought of.
- These inventions are called frugal inventions.
- In India there is a culture of repairing, which is not present in the western countries.
- In fact there are even laws against repairing some parts.
- Eg: Exhaustive use of vehicles for 30-35 years

#Entrepreneurship

Scaling the innovation to solve the problem of masses

Etymology:

To do something that leads to revenue.

Entrepreneur:

A person who sets up a business/es, taking on financial risks in the hope of profit

Enterprise:

- A large scale organization engaged in commercial activities, has complex network of operations, departments, and divisions.
- Entity that the entrepreneur built

Roles of an Entrepreneur:

(i) To solve a problem:

- should solve a problem of the masses

(ii) Be responsible:

- should be responsible on behalf of all works, employees, etc as the are a responsibility of an entrepreneur.

(iii) Be independent:

- An entrepreneur is independent, they work as they will for the benefit of the enterprise keeping in mind laws, regulations and ethics.

(iv) Wealth generation and distribution:

- It is a responsibility of an entrepreneur that their enterprise generates revenue and people working under them grow with the enterprise, hence wealth distribution.

(v) Taking care of employees:

- Entrepreneur must make sure that the people grow as well
- It is their responsibility
- A huge role is not only be independent, but also to make people independent.

Eg: AMUL has made many women independent

Intrapreneur

A manager that works within the company who promotes innovative product development and marketing.

Eg: Sundar Pichai → joined company and raised to CEO

7 I's of Intrapreneur:

(i) Innovation:

- Modifying of an existing technology / Invention.

(ii) Intuitiveness:

- Ability to foresee trends, have good instincts, which help in making good decisions

(iii) Integrator:

- Connect different ideas, teams, technologies to create seamless solutions.

(iv) Initiator:

- Take actions on their own than waiting for decision

(v) Independence:

- Even in a company, intrapreneurs take their own decision

(vi) Influencer:

- Inspire and persuade others to support their idea

(vii) Insider:

- Have a good knowledge of the operations, culture, and goals of the company so that they can align their ideas with business objectives

Stages of Entrepreneurship:

1) **Discovery** of Idea / Problem / Issue

2) **Conceptual Development** (Build up on that idea)

→ Take advice

→ Brainstorming

→ Why the issue (causes)

→ Full fledged plan

→ Come up with solutions

→ Pick the best solution

3) **Resourcing**: Human resources, technical resources, financial resources
→ Hire people, arrange funds, get investors

(4) Actualization:

Gather resources and build the enterprise

(5) Harvesting:

Make profit, yield end product of the result

Keep your assets to grow further

#Types of Enterprise

- | | |
|--------------------|-------------------|
| (i) Product based | (iii) Urban based |
| (ii) Service based | (iv) Rural based |

Case Study:

→ Mansukh Bhai Prajapati (Potten)

→ Mitti cool

- Demand for clay pottery reduced
- He had failed 10th class
- He came up with a refrigerator made up of clay, which was free from electricity consumption
- His spark for this idea was due to a news photo which was captioned "Gharibi Ka Fridge Tut Gaya"
- He started experimenting different types of clays, sands and sawdust to make a porous soil that kept things cool

#Types of Businesses

1) Small scale business:

- Starts with limited resources, operates on small scale and expands over time
- Eg: Rameshwara Café, Café Coffee Day, Sun Entertainment, etc

2) Trading Entrepreneurship:

- Buying of goods and selling to another, basically middleman.
- Eg: Wholesale and retail, import-export firms

3) Industrial Entrepreneur:

- Involved in manufacturing or production
- Large scale operation
- Eg: Al Anwar Ceramic Tiles Company (Al Shams), Raymond

4) Corporate Entrepreneurship

- Large corporation that focus on innovation, expansion or creation of new business ventures.
- Eg: Reliance Industries, Tata group

5) Agricultural Entrepreneur:

- That undertake agricultural activities such as dairy, horticulture marketing of crops, fertilizers
- Eg: Amul, Hattsum

6) Retail

- Activities such as supermarket, shopping malls, teleshopping, etc
- Eg: Walmart, Flipkart

7) Service

- Provide services
- Eg: Apollo Hospital, Catering services

8) Urban / Rural

- Business that caters to urban markets or rural markets
- Eg: King Fisher Airlines, Retailing

9) Technical

- Establish technology related enterprises
- Eg: Bose, Infosys

10) Professional entrepreneur

- Make product → sell product → move on
- Then start with another business
- Eg: ITC

11) Large Scale Entrepreneur

- Start a business with large capital investment and employs large number of people with high output
- Eg: Aditya Birla Group

Concept of Creativity

↳ New ideas

↳ Create something new (out of the box)

Eg: Dettol vs Savlon

↳ without burning sensation

Eg: flavours of drinks

Eg: Big basket (fast delivery)

Stages of Creativity:

1) Preparation:

- Formal education
- Work experience
- Discuss with experts
- Brainstorming sessions
- Seminars and lectures

3) Transformation

- Visualize no. of approaches to a problem

4) Incubation:

- Study the problem
- Give time to reflect on information

2) Investigation:

- Facts
- Memory
- Imagination

5) Illumination:

- Screen ideas to the problem
- Find invention / discover new things

6) Verification:

- Piloting
- Test marketing
- Simulation
- Prototype building
- Various experiments

7) Implementation:

- Process of bringing idea into reality

Sources of new ideas

- ① **Consumers**: (Eg: Mitti cool, idea from news caption that read "Gharibi ka Fridge Tut Gaya") (Eg: Paytm)
- ② **Competitors**: Evaluate others products in the market which may spark a new idea. (Eg: Jio's entry with free / affordable plans)
- ③ **Distribution channels**: members of the distribution channels know the needs of people, hence can prove a great source of idea. (Eg: Flipkart's kirana network)
- ④ **Government**: (i) Patent office files contain info on product
(ii) New government regulation can give a new prod idea
(Eg: Startup India Initiative)
- ⑤ **R & D**: (Eg: Mahindra) Research can spark new ideas
(Eg: ISRO's low cost Mars mission)
- ⑥ **Demography**: (Eg: Fast fashion Zudio)
 - when the economy is unstable
 - market and industry changes
 - Eg: Rise of Byju's and Unacademy due to large youth population and ease of access of the internet.

6 Hat Theory:

- Edward de Bono's Six thinking hat theory
- A tool for parallel thinking, decision making, and problem solving.

(i) White Hat (Purity) ("Facts and Information"):

- Factual Hat (gather information about different kinds of problems)
- Epistemology (acquire knowledge)
- Focuses on objective data, facts and statistics.
- Eg: Business meetings to analyze sales numbers, market status, etc.

(ii) Red Hat ("Emotions and Feelings"):

- Represents intuition, emotions and gut feelings
- Eg: Human relationships are assessed based on emotions

(iii) Black Hat (Critical judgement and Caution):

- Doom, risk, weakness and potential problems
- Eg: Before expanding a business, or taking up an offer, we are skeptical, or think about risks, etc.

(iv) Yellow Hat (Optimism and Benefits)

- positive aspects, opportunities, and advantages
- Eg: Company wants to enter a new market, yellow hat would highlight the positive aspects, profit, etc.

(v) Green Hat (Creativity and New ideas)

- Innovation, brainstorming, thinking outside the box
- Eg: Web designer making the design of a website, etc.

(vi) Blue Hat (Process and Control):

- Organizing, planning, managing
- Eg: Project manager make sure to manage the team.

Howard Gardner's Theory of Intelligence

- Theory of Multiple intelligence
- Not just IQ (single measure)
- It is multifaceted (has many sides)

Initially proposed 8 intelligence:

1) Logical - Mathematical Intelligence (Number / Reason Smart)

- problem solving skills
- logical reasoning
- numerical ability
- Eg: Mathematicians, Scientists, etc

2) Bodily - Kinesthetic Intelligence (Body Smart)

- Control body movements
- Handle objects skillfully
- Eg: Athletes, surgeons, etc

3) Musical Intelligence (Music Smart)

- Sensitive to sound pattern, rhythms, tones
- Ability to compose or appreciate music
- Eg: Musicians, singers, etc

4) Linguistic Intelligence (Word Smart)

- Use language effectively
- Communicate, write, read, and storytelling
- Eg: Writers, poets, speakers, even lawyers.

5) Spatial Intelligence (Picture Smart)

- Ability to visualize, manipulate, and interpret images and spatial relationships.
- Eg: Architects, designers, artists

6) Interpersonal Intelligence (People Smart)

- Capacity to interact, understand and communicate well with others.
- Eg: Teachers, therapist, politicians, etc

7) Intrapersonal Intelligence (Self Smart):

- Ability to understand ones emotions, motivations and inner thoughts
- Eg: Self aware people, psychologist, philosophers, etc

8) Naturalistic Intelligence (Nature Smart):

- Ability to identify, classify and engage with the natural world.
- Eg: Biologist, farmers, etc.

Later Howard Gardner also suggested a possible 9th intelligence although he has not added this officially to the list of multiple intelligence.

9) Existential Intelligence (Big Picture Smart)

- Ability to think and understand questions about human existence and purpose of life
- Eg: Philosophers, spiritual leaders, etc.

Types of Innovations

1) Technical innovation:

- Modifying / Updating existing products, process or services with the help of new technology.
- Eg: 5G in India, AI powered chatbots, etc

2) Organizational Innovation:

- Restructuring the workflow, or management process in an organization
- Eg: Covid 19 introduced work from home, Agile project management

3) Product Innovation:

- Creation or improvement of a product
- Eg: Amul's invention of flavoured milk and lactose-free dairy Samsung flip and fold product lines.

4) Radical Innovation:

- A completely new breakthrough that transforms industries or creates new markets.
- Eg: AI driven automation, Tesla's self driving car, etc

5) Marketing Innovation:

- A new way / strategy for promoting product and services
- Eg: Flipkart Big Billion Days campaign, Coca-Cola's personalized name bottles

Property:

- a thing or things that belong to someone
- Eg: Land, flat, etc

Intellectual Property Rights (IPR)

- Refers to legal protections granted to creators and inventors for their intellectual works.
- Allows / Permits control, profit from and protect their innovation, creative works, and brands.

Patents:

- An exclusive legal right granted to an inventor for a new invention, allowing them to prevent others from using, making or selling the invention without permission.
- Granted for a fixed period of time after which it enters public domain (usually 20 years)

#Types of Patents:

1) Utility Patent:

- Functionality / working mechanism of a product.
- Validity: 20 years from the filing date
- Eg: Amazon's "One-Click Purchase" system, Apple's multitouch gesture iPhone

2) Design Patent:

- Aesthetic design
- Validity: 10 years (+5 years extended in India)
- Eg: Coca-Cola bottle shape, Apple's iPhone design, etc

3) Plant Patent:

- For new varieties of plants that are asexually produced
- Validity: 20 years
- Eg: BT cotton, Alphonso Mango Hybrid

4) Product Patent

- Protects end product
- Eg: iPhone processors, Pfizer's Covid-19 vaccine, etc

5) Process Patent:

- Process or method to make a product rather than the final product.
- Eg: Coca-Cola's secret formula, etc